AP PRECALCULUS

Unit 1 • Lesson 1.5 practice

Lesson 1.5 • Polynomial Functions and Complex Zeros

4 multiple-choice questions and 1 free-response question

Question 1

How many complex zeros, counted with multiplicity, does a polynomial of degree 6 have?

(A) 3
(B) 5
(C) 6
(D) 12

Question 2

What are the zeros of $p(x) = x^3 + 4x$?

(A) $0,\ 2,\ -2$
(B) $0,\ 2i,\ -2i$
(C) $0,\ 4i$
(D) 0 only

Question 3

A polynomial with real coefficients has zeros 3 and $2i$. What is its least possible degree?

(A) 2
(B) 3
(C) 4
(D) 6

Question 4

Which function is odd?

(A) $f(x) = x^4 + x$
(B) $f(x) = x^3 + 1$
(C) $f(x) = x^3 - 5x$
(D) $f(x) = |x|$

Question 5 (Free response)

Let $p(x) = (x-1)^2(x+3)(x^2+1)$.

(a) Find the degree of p and all of its complex zeros with their multiplicities.
(b) Describe the behavior of the graph of p at each x-intercept.
(c) Is p even, odd, or neither? Justify your answer.